

Supplementary material: HCA-DSE

Full budget sweeps and validation subsets accompanying the main manuscript. All values are generated from the same result CSVs as the main-text summaries.

Table S1: Full budget-matched comparison on S2. Medians over 20 seeds.							
scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S2	telemetry	exhaustive	3 888	1.000	1.000	0.0000	0.045
		random ($B=100$)	100	0.037	0.692	0.1496	0.001
		random ($B=250$)	250	0.037	0.794	0.1018	0.003
		random ($B=500$)	500	0.074	0.880	0.0713	0.007
		random ($B=1000$)	1000	0.185	0.923	0.0451	0.014
		NSGA-II ($B=100$)	100	0.037	0.703	0.1537	0.011
		NSGA-II ($B=250$)	250	0.259	0.886	0.0447	0.029
		NSGA-II ($B=500$)	500	0.759	0.995	0.0063	0.072
		NSGA-II ($B=1000$)	1000	0.944	0.999	0.0003	0.148
		HCA-DSE	46	1.000	1.000	0.0000	0.013
S2	control	exhaustive	3 888	1.000	1.000	0.0000	0.049
		random ($B=100$)	100	0.040	0.691	0.1460	0.001
		random ($B=250$)	250	0.040	0.792	0.0940	0.003
		random ($B=500$)	500	0.080	0.879	0.0566	0.006
		random ($B=1000$)	1000	0.160	0.923	0.0379	0.014
		NSGA-II ($B=100$)	100	0.040	0.627	0.1592	0.011
		NSGA-II ($B=250$)	250	0.300	0.910	0.0348	0.030
		NSGA-II ($B=500$)	500	0.800	0.991	0.0041	0.072
		NSGA-II ($B=1000$)	1000	0.900	0.999	0.0007	0.161
		HCA-DSE	39	1.000	1.000	0.0000	0.014
S2	sensing	exhaustive	3 888	1.000	1.000	0.0000	0.029
		random ($B=100$)	100	0.000	0.469	0.2629	0.001
		random ($B=250$)	250	0.062	0.686	0.1521	0.002
		random ($B=500$)	500	0.062	0.804	0.0808	0.005
		random ($B=1000$)	1000	0.188	0.866	0.0622	0.010
		NSGA-II ($B=100$)	100	0.000	0.404	0.3482	0.010
		NSGA-II ($B=250$)	250	0.312	0.793	0.0782	0.031
		NSGA-II ($B=500$)	500	0.719	0.965	0.0224	0.077
		NSGA-II ($B=1000$)	1000	0.938	1.000	0.0009	0.186
		HCA-DSE	26	1.000	1.000	0.0000	0.010

Table S2: Full budget-matched comparison on S3. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S3	telemetry	exhaustive	34 560	1.000	1.000	0.0000	0.320
		random ($B=100$)	100	0.000	0.259	0.3903	0.001
		random ($B=250$)	250	0.000	0.487	0.2574	0.002
		random ($B=500$)	500	0.000	0.563	0.1671	0.005
		random ($B=1000$)	1000	0.016	0.699	0.1147	0.011
		NSGA-II ($B=100$)	100	0.000	0.151	0.4919	0.009
		NSGA-II ($B=250$)	250	0.000	0.687	0.1732	0.026
		NSGA-II ($B=500$)	500	0.355	0.915	0.0558	0.061
		NSGA-II ($B=1000$)	1000	0.645	0.987	0.0415	0.146
		HCA-DSE	52	1.000	1.000	0.0000	0.055
S3	control	exhaustive	34 560	1.000	1.000	0.0000	0.276
		random ($B=100$)	100	0.000	0.262	0.4192	0.001
		random ($B=250$)	250	0.000	0.492	0.2537	0.002
		random ($B=500$)	500	0.000	0.564	0.1746	0.005
		random ($B=1000$)	1000	0.018	0.691	0.1248	0.010
		NSGA-II ($B=100$)	100	0.000	0.286	0.4405	0.010
		NSGA-II ($B=250$)	250	0.037	0.748	0.1009	0.025
		NSGA-II ($B=500$)	500	0.278	0.933	0.0387	0.063
		NSGA-II ($B=1000$)	1000	0.648	0.991	0.0244	0.147
		HCA-DSE	43	1.000	1.000	0.0000	0.053
S3	sensing	exhaustive	34 560	1.000	1.000	0.0000	0.282
		random ($B=100$)	100	0.000	0.037	0.6082	0.001
		random ($B=250$)	250	0.000	0.231	0.3843	0.002
		random ($B=500$)	500	0.000	0.556	0.2543	0.005
		random ($B=1000$)	1000	0.000	0.600	0.1747	0.010
		NSGA-II ($B=100$)	100	0.000	0.050	0.5191	0.011
		NSGA-II ($B=250$)	250	0.037	0.279	0.2833	0.025
		NSGA-II ($B=500$)	500	0.370	0.750	0.1996	0.064
		NSGA-II ($B=1000$)	1000	0.741	0.969	0.1670	0.165
		HCA-DSE	43	1.000	1.000	0.0000	0.071

Table S3: Full S3 statistical comparison: one-sided Mann–Whitney test for larger NSGA-II hypervolume and Cliff’s δ . Unadjusted p -values.

workload	budget	NSGA-II HV med.	random HV med.	p	Cliff’s δ	NSGA-II recall	HCA-DSE evals
telemetry	100	0.1509	0.259	0.951	-0.302	0.0	52
telemetry	250	0.6868	0.4871	0.021	0.38	0.0	52
telemetry	500	0.9154	0.5633	<0.001	0.7	0.355	52
telemetry	1000	0.9874	0.6993	<0.001	0.7	0.645	52
control	100	0.2859	0.2616	0.484	0.01	0.0	43
control	250	0.7481	0.4916	<0.001	0.62	0.037	43
control	500	0.9331	0.5644	<0.001	0.8	0.278	43
control	1000	0.9906	0.6913	<0.001	0.8	0.648	43
sensing	100	0.0497	0.0367	0.261	0.12	0.0	43
sensing	250	0.2787	0.2308	0.672	-0.08	0.037	43
sensing	500	0.7504	0.5561	0.168	0.18	0.37	43
sensing	1000	0.9689	0.5998	0.054	0.3	0.741	43

Table S4: Testbed validation against measured p_{50} , including utilisation and reproducibility subsets. Pair agreement is a fraction; undefined values (no eligible pairs) are shown as –.

subset	n	MAPE [%]	med. AE [ms]	Spearman ρ	Kendall τ	pair > 10%	pair > 25%	repeat IQR [ms]
telemetry (all)	12	19.9	7.43	0.811	0.697	0.815	0.895	0.23
telemetry ($\rho \leq 0.75$)	6	15.4	2.85	0.829	0.733	0.818	1.0	0.2
control (all)	12	14.4	2.49	0.865	0.748	0.896	0.909	0.11
control ($\rho \leq 0.75$)	10	9.4	2.32	0.766	0.629	0.815	0.833	0.11
sensing (all)	12	51.1	19.15	0.935	0.84	0.935	0.913	1.0
sensing ($\rho \leq 0.75$)	10	51.0	18.82	0.985	0.944	1.0	1.0	0.75
sensing (reproducible)	8	39.3	18.56	0.97	0.909	1.0	–	0.67
all workloads (all)	36	28.4	7.43	0.968	0.884	0.961	0.98	0.27
all workloads ($\rho \leq 0.75$)	26	26.8	3.96	0.945	0.855	0.959	0.982	0.22
all workloads (reproducible)	32	22.7	4.5	0.957	0.869	0.956	0.979	0.23

Table S5: Full order/queue ablation on S1–S3. All configurations preserve the unrounded exhaustive front.

scale	workload	order	queue	states	checks	evals	active	pruned	ms
S1	telemetry	determinants-early	off	447	746	136	0	0	6.7
S1	telemetry	determinants-early	on	249	516	32	306	46	5.2
S1	telemetry	determinants-late	off	695	1299	136	0	0	9.5
S1	telemetry	determinants-late	on	589	1299	30	0	0	10.3
S1	control	determinants-early	off	391	669	113	0	0	5.2
S1	control	determinants-early	on	242	512	28	312	35	4.6
S1	control	determinants-late	off	679	1317	112	0	0	9.6
S1	control	determinants-late	on	593	1317	26	0	0	10.5
S1	sensing	determinants-early	off	481	810	147	0	0	5.8
S1	sensing	determinants-early	on	226	503	16	309	72	4.1
S1	sensing	determinants-late	off	497	819	148	0	0	6.1
S1	sensing	determinants-late	on	365	819	16	0	0	5.8
S2	telemetry	determinants-early	off	939	1815	208	0	0	15.4
S2	telemetry	determinants-early	on	604	1383	46	839	99	13.9
S2	telemetry	determinants-late	off	1816	3940	208	0	0	30.7
S2	telemetry	determinants-late	on	1652	3940	44	0	0	35.8
S2	control	determinants-early	off	835	1674	165	0	0	13.9
S2	control	determinants-early	on	587	1364	39	842	80	12.0
S2	control	determinants-late	off	1780	3960	164	0	0	29.0
S2	control	determinants-late	on	1653	3960	37	0	0	32.9
S2	sensing	determinants-early	off	1161	2184	289	0	0	15.0
S2	sensing	determinants-early	on	600	1427	26	879	177	10.1
S2	sensing	determinants-late	off	1268	2421	290	0	0	16.8
S2	sensing	determinants-late	on	1004	2421	26	0	0	15.8
S3	telemetry	determinants-early	off	2871	6806	251	0	0	53.3
S3	telemetry	determinants-early	on	2397	6100	52	3777	188	54.4
S3	telemetry	determinants-late	off	6029	14794	251	0	0	117.9
S3	telemetry	determinants-late	on	5828	14794	50	0	0	131.9
S3	control	determinants-early	off	2755	6636	206	0	0	51.0
S3	control	determinants-early	on	2375	6074	43	3777	163	56.3
S3	control	determinants-late	off	5990	14812	205	0	0	106.7
S3	control	determinants-late	on	5826	14812	41	0	0	127.5
S3	sensing	determinants-early	off	4351	9905	537	0	0	72.9
S3	sensing	determinants-early	on	3221	8281	43	5139	419	60.9
S3	sensing	determinants-late	off	4888	11367	538	0	0	82.7
S3	sensing	determinants-late	on	4393	11367	43	0	0	82.3